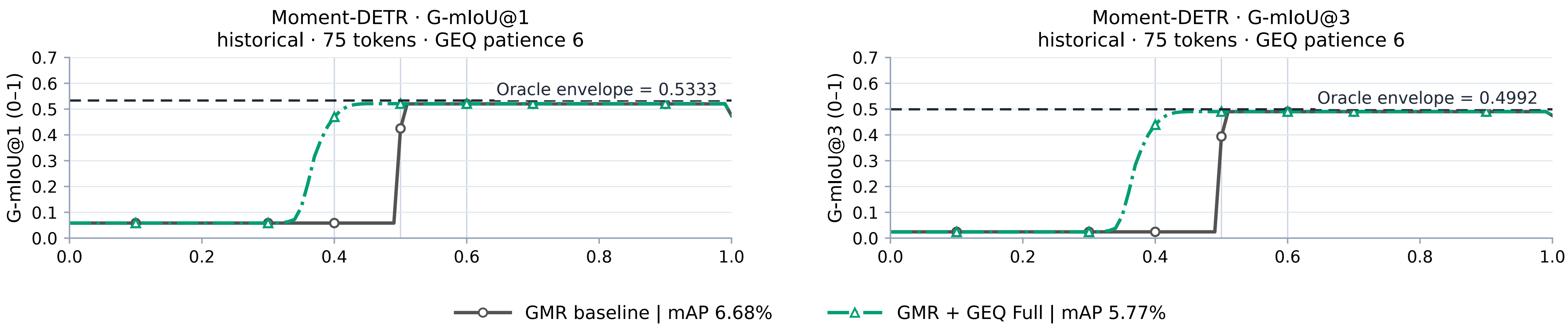
11 backbones · GMR baseline vs GMR + GEQ Full

Standard-trained, fixed Val-selected checkpoints → SoccerGMR Standard Test (1036 queries)

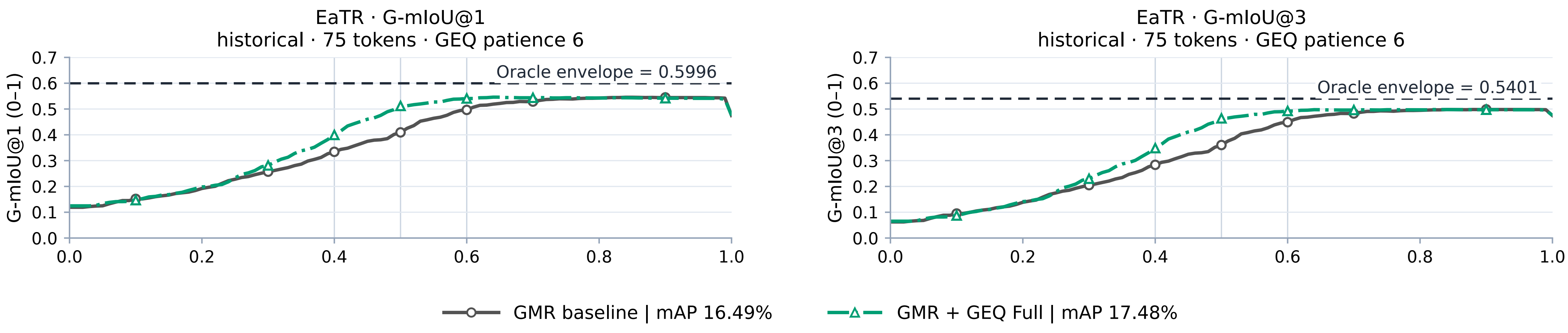
Historical 5: GEQ patience 6, video limit 75 | Expansion 6: GEQ patience 10, video limit 128, seed 2023



ΔmAP: -0.91 pp

ΔG1 mean: +16.95 pp

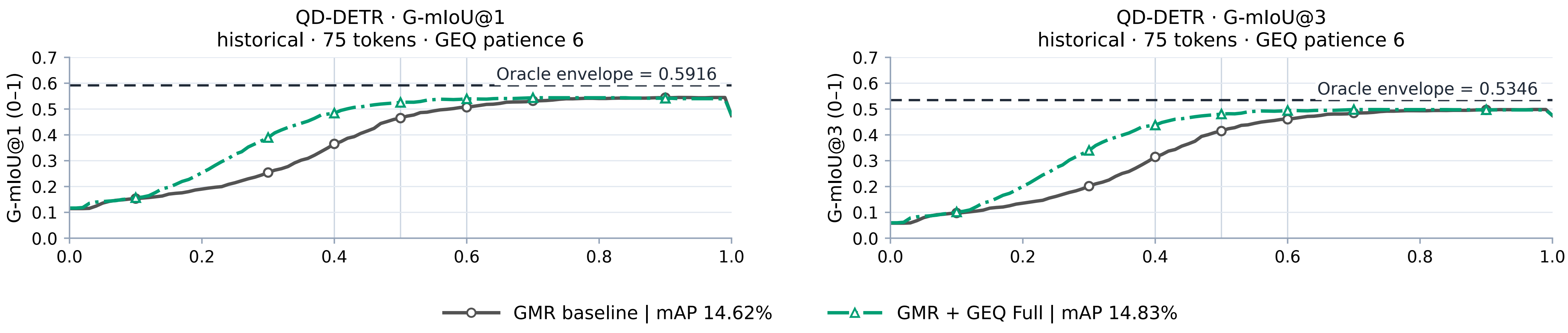
ΔG3 mean: +17.06 pp



ΔmAP: +0.99 pp

ΔG1 mean: +7.07 pp

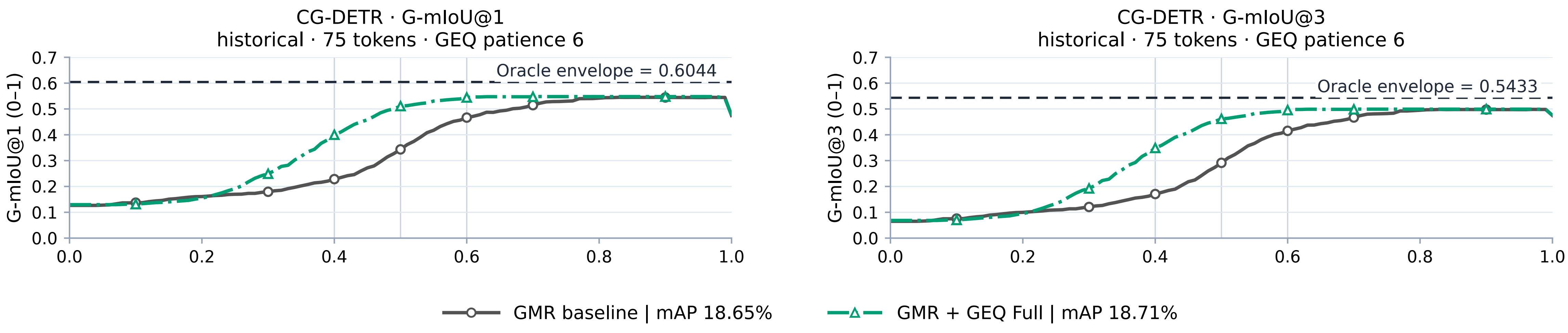
ΔG3 mean: +7.05 pp



ΔmAP: +0.21 pp

ΔG1 mean: +7.08 pp

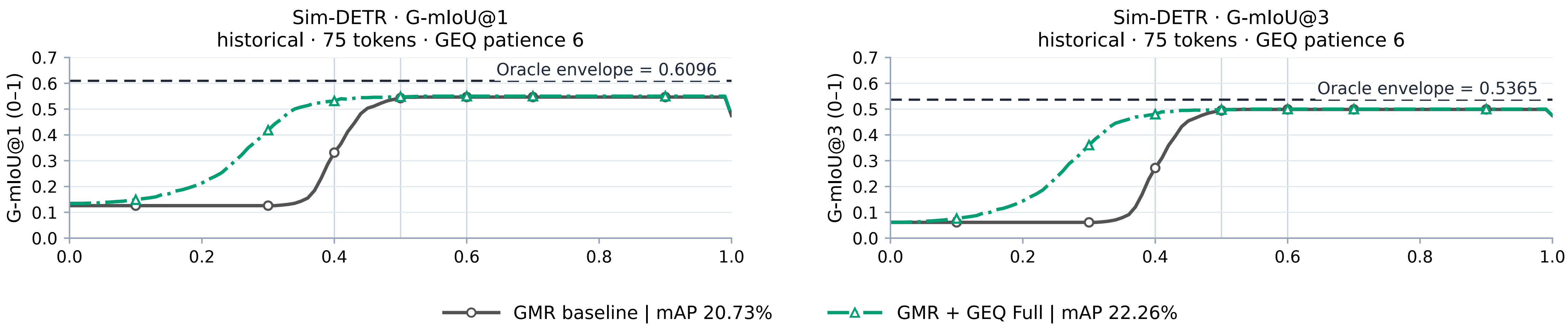
ΔG3 mean: +7.44 pp



ΔmAP: +0.06 pp

ΔG1 mean: +13.90 pp

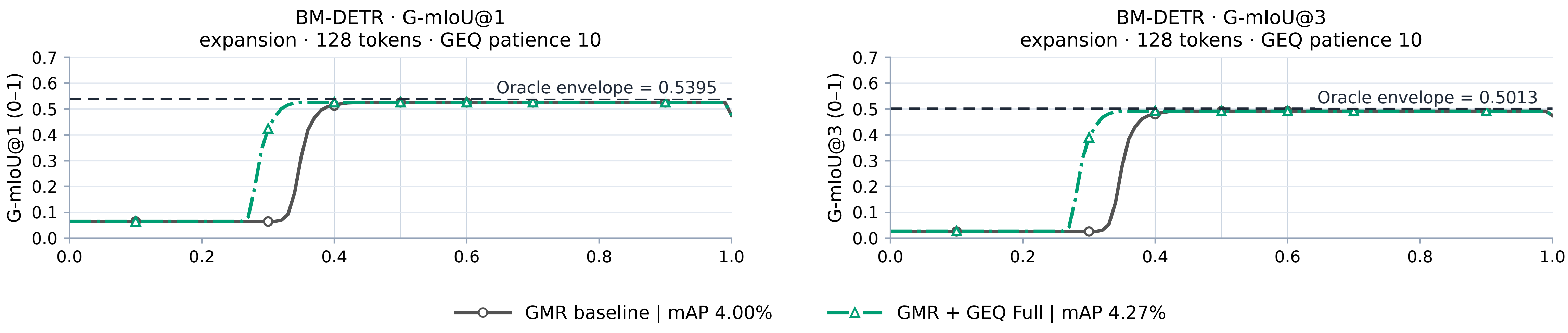
ΔG3 mean: +14.30 pp



ΔmAP: +1.53 pp

ΔG1 mean: +6.97 pp

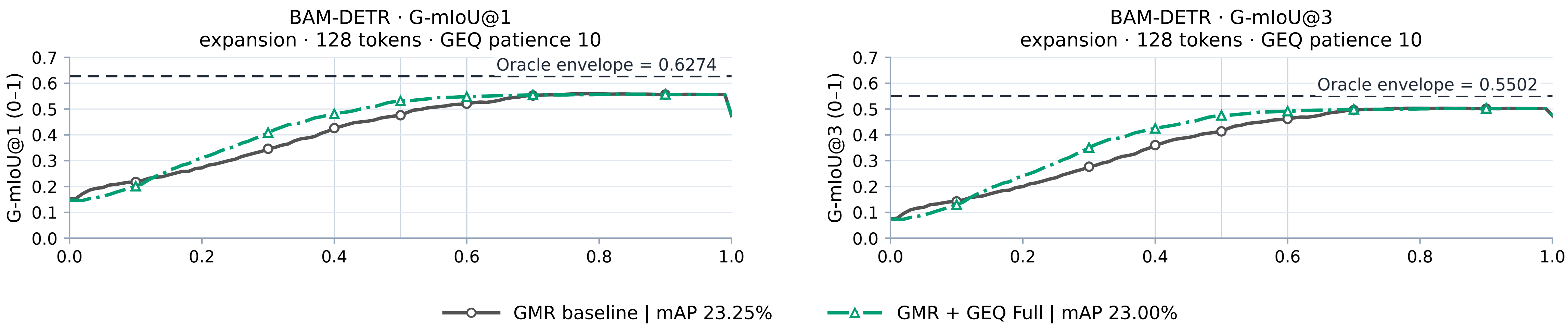
ΔG3 mean: +7.11 pp



ΔmAP: +0.27 pp

ΔG1 mean: +0.44 pp

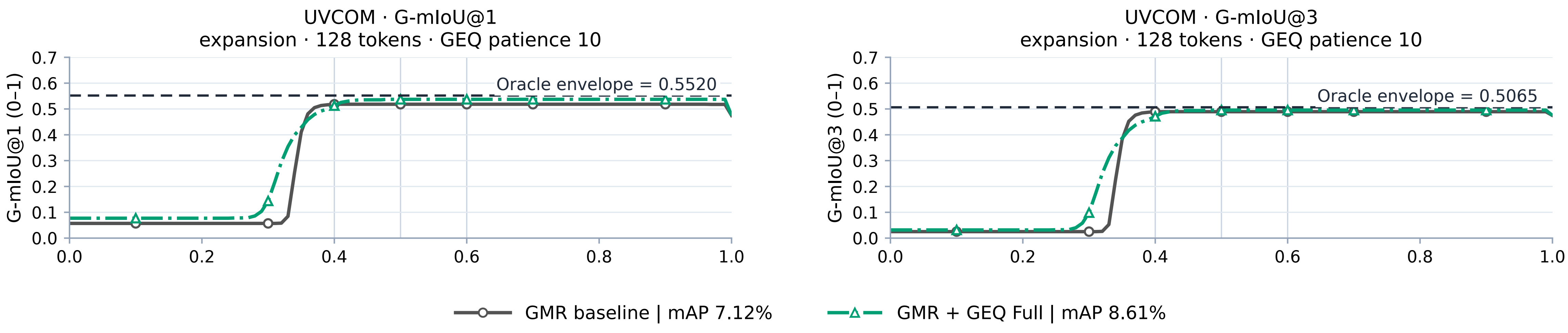
ΔG3 mean: +0.42 pp



ΔmAP: -0.25 pp

ΔG1 mean: +4.56 pp

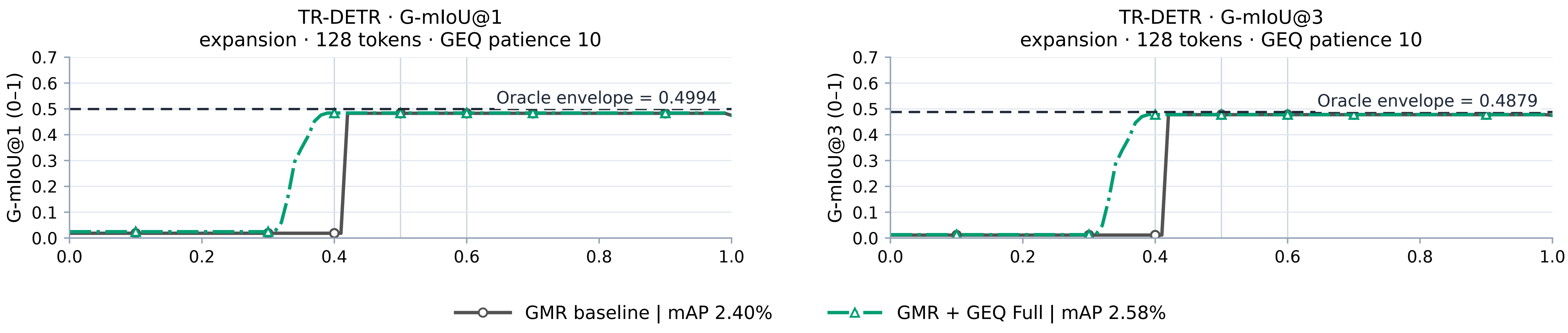
ΔG3 mean: +5.20 pp



ΔmAP: +1.49 pp

ΔG1 mean: +1.09 pp

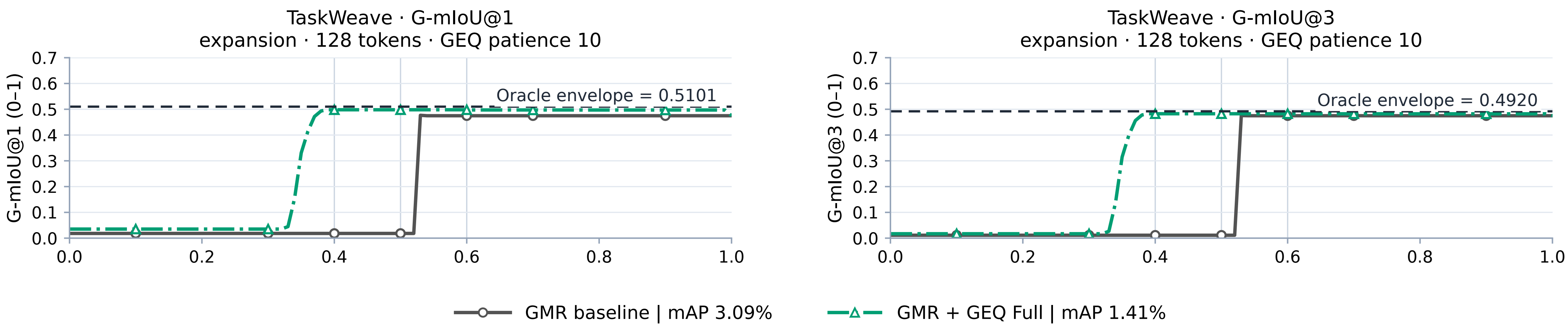
ΔG3 mean: -0.18 pp



ΔmAP: +0.18 pp

ΔG1 mean: +15.60 pp

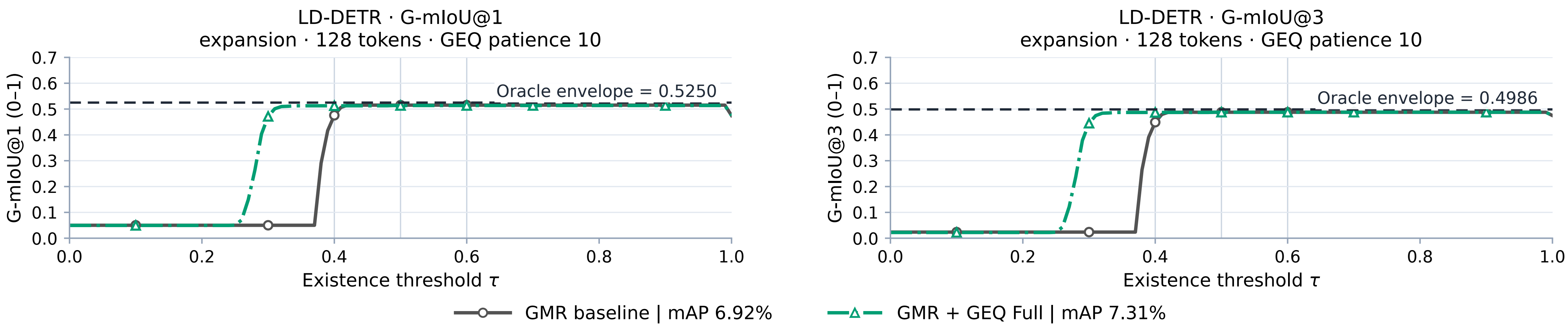
ΔG3 mean: +15.58 pp



ΔmAP: -1.68 pp

ΔG1 mean: +32.75 pp

ΔG3 mean: +31.68 pp



ΔmAP: +0.39 pp

ΔG1 mean: +1.13 pp

ΔG3 mean: +1.27 pp

Curves: 101 fixed thresholds, 0–1 units. Δ = GEQ Full − matching GMR; G1/G3 means use τ = .4/.5/.6; pp = percentage points.

Oracle envelope: GT presence with each method’s fixed candidates; diagnostic only. Test never selects checkpoints or thresholds.

One seed per experiment; curves show threshold sensitivity, not run uncertainty. Protocol groups are not a controlled cross-backbone ranking.